

operations are going to require a decent amount of processor power. So if you have a processor which isn't fast, virtualization will make you feel the pain even more so. At times, you may encounter freezes of the Host operating system. Now, when we say a processor that's "powerful enough", we're not simply meaning the "clock frequency". So if your neighbor has a dual core Core-i3 processor which runs at 1.87 GHz per core and you have an Atom processor which runs at the same speed, your performance is going to be low. If there's a resounding "why" in your mind, you should know that the performance of the processor not only depends on the clock frequency but also on a couple of other major factors namely the "Supported Instruction Set" and "Cache Size". Since a processor belonging to "Atom" line won't be having as much Cache as a processor belonging to the Core-i3 line, Atom would go slow.

Hardware restriction such as "heat-dissipation" is another factor for the performance bottleneck (which indirectly means that a desktop computer with average cooling is going to be better than a laptop with the same specifications (well, yes the processors would vary in the "model" but the "specs" can be the same). You're probably convinced by now that virtualization is going to require a decent amount of processing power, although it doesn't demand a high performance rig either. A laptop would work just as well for virtualization as long as you take care of cooling.

For those who are familiar with the OS architecture (engineering students, eh?) you probably know about the ring structure of the operating system wherein the core system functionality is actually run on the processor in a "privileged" mode while the rest is running in a "non-privileged" mode. In case of virtualization, wherein you're trying to emulate a "processor", it's necessary that the processor is emulated well. This emulation requires massive processor power when the virtualization software is trying to emulate the "privileged" and "non-privileged" modes of the processor to the guest operating system. The reasons are obvious – the virtualization software itself runs in "user" mode or "non-privileged" mode. When the guest operating system issues a hardware request (such as a "disk read"), it has to emulate such events and they must look like a "privileged mode" operation to the guest. Hence, overall, the scene is such: a "non-privileged" process trying to execute "privileged" code. Such emulation is really costly on the processor and takes up plenty of processing power as the virtualization software tries to switch to and from the "privileged" and "non-privileged" modes for the guest while asking the host operating system for resources.